Golden Config

AI Agent – Project

Infinite Computer Solutions

Problem Statement: Recurring Configuration issues

Config outages
80%

80% Unplanned Outages are due to config issues + poor change Management

- *IT Process Institute (Ops Handbook)*

Annual Impact
-$115B

Downtime and Bug fix losses

- *US Industry data (IDC Survey)*

Use case: The Adapter Service Defect

- **What Happened ?:** A developer, working on adapter service made a configuration change.
- **The Defect:** The .yaml file was incorrectly updated to point to a dev URL instead of correct URL (ptg-alpa).
- **The Impact:** This change was merged and deployed , breaking the entire non-prod environment
- **Root Cause:** This is a “**contextual error”** looked valid but was disastrous and hard to catch

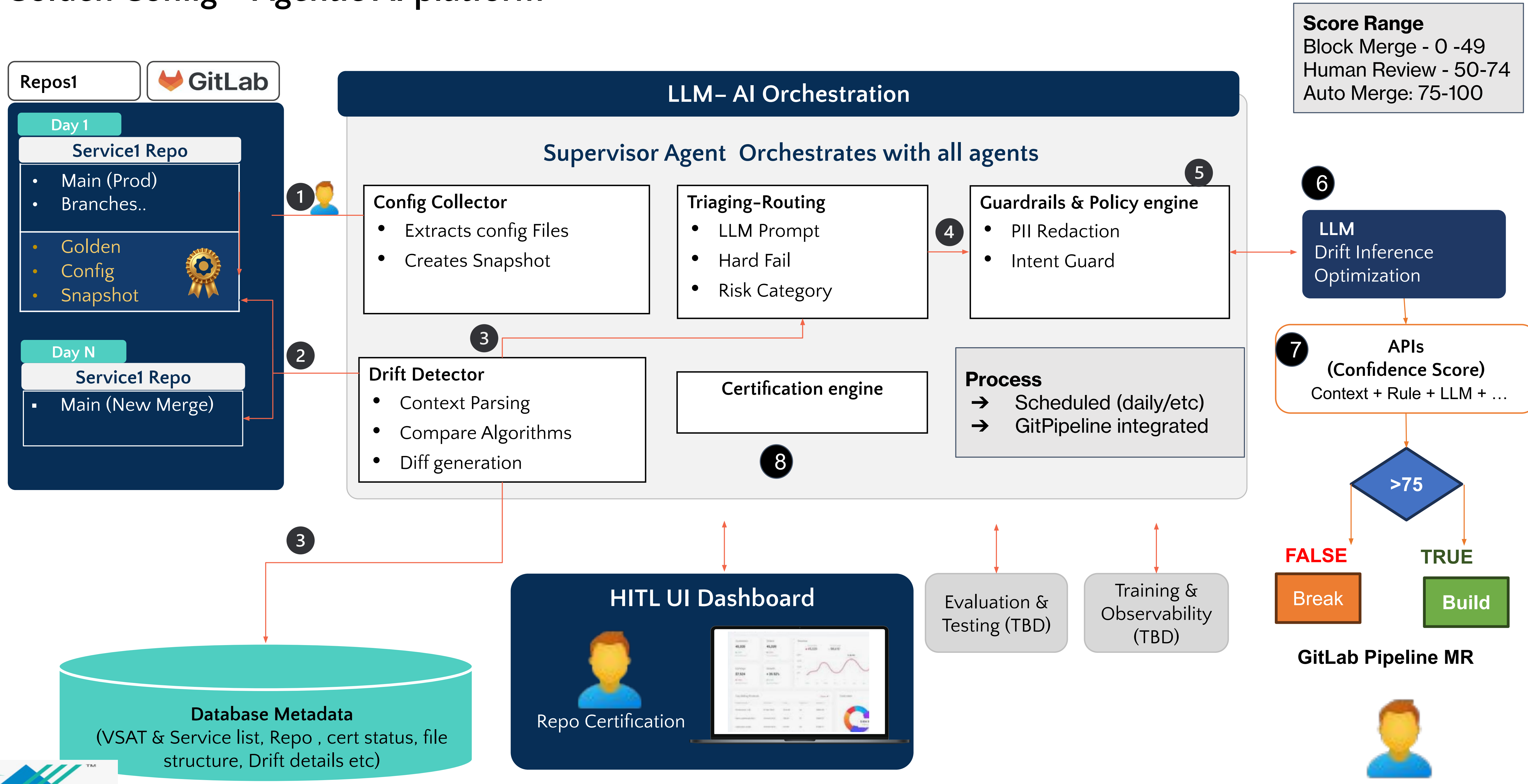
Wrong Service: dev-alpha

Correct Service: ptg-alpha

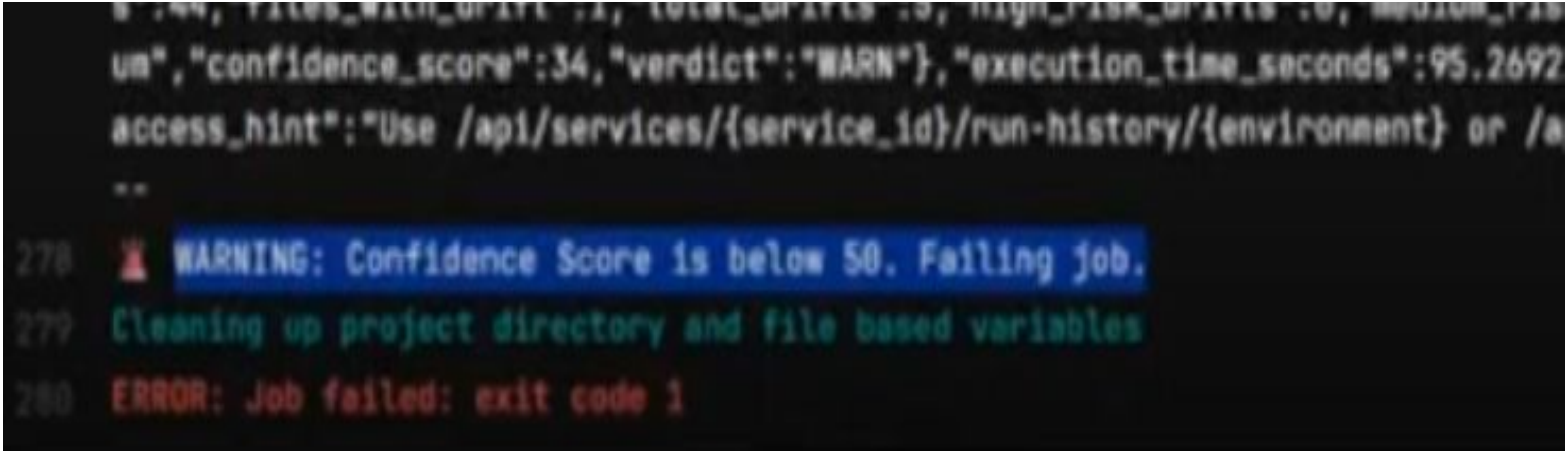
How AI Integrates:

- **Automated Drift detection (Proactive)**
- **Blocking Git Merge on Confidence score**
- **Audit Trails and Compliance + more …**

Golden Config – Agentic AI platform



GIT Pipeline Integration - API confidence score



Score Breakdown - PTG Adapter

LLM

Source	Details	Score
Baseline	Start at full confidence	**100**
Penalty (High)	**Service ID not in mismatch Beta2 (unknown alias). Deterministic hard rule hit.**	**−40**
Impact Penalty - Magnitude	**5 drifted references in a critical adapter YAML (single change fan-out, high blast radius).**	**−25**
HistoryScore	Prior alias/route typo caused outage in this area → slight distrust.	**−5**
LLM_SafeProb	LLM sees ptg-sqa3 used many times, 08v-beta2 never; infers likely typo → very low safety (scaled).	**4**
ContextBonus	No MR tag like [rename-adapter], no Jira rename plan, no rollback note.	**0**
Final Confidence	100 − 40 − 25 − 5 + 4 + 0	**= 34 → BLOCK**

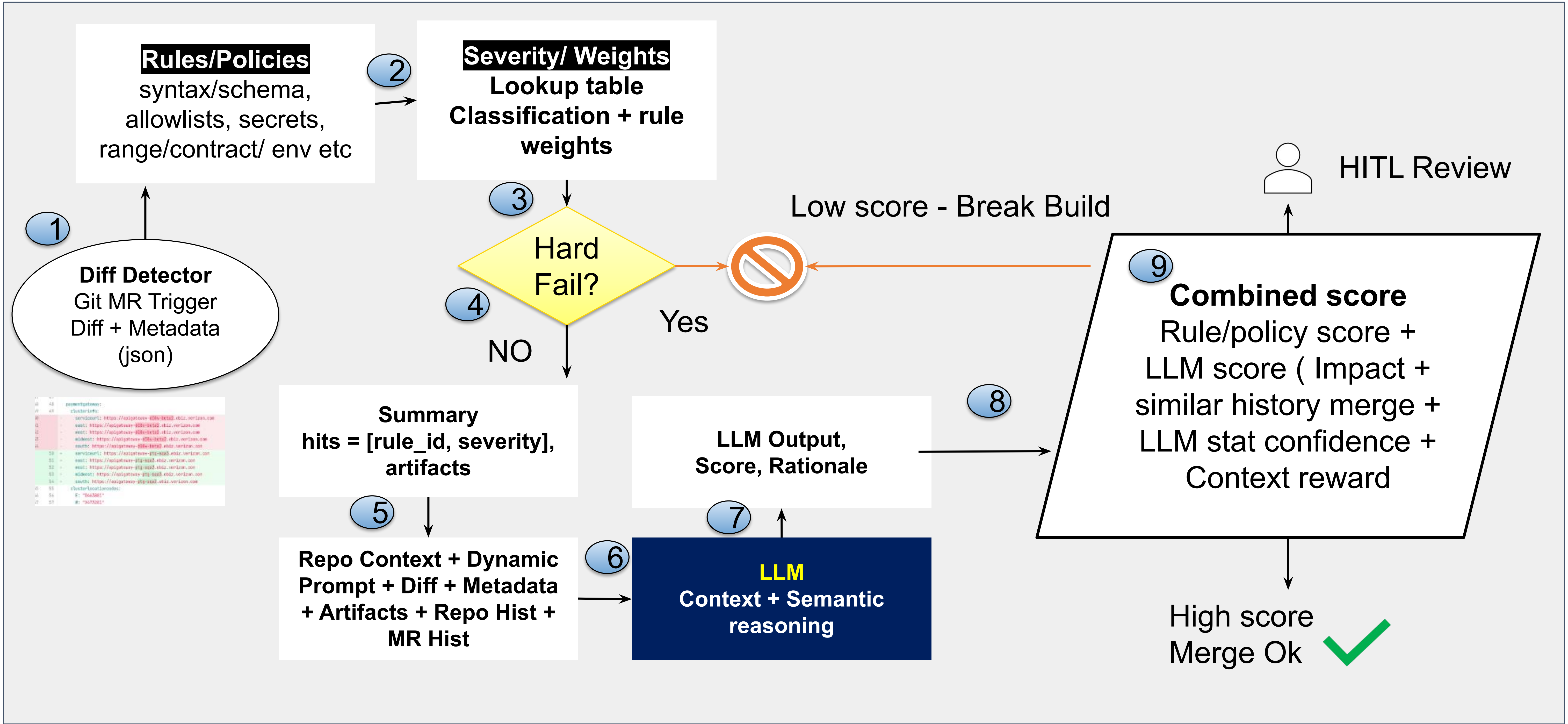
Production Build

Business: We will be integrating all the Repos

1. Integrate to all non-prod services - Deploy Production-Grade Infrastructure
2. Integrate with "Guardian UI", Pipeline Integration (Like SonarQube)
3. Develop Scalable, Modular Agents
4. Build Centralized Postgres Database
5. Implement Comprehensive Quality Assurance, MR Threshold
6. AI Automation Pipeline
7. Implement Skip governance where AI check is not needed.

Decision & Scoring Flow

Score Range
Block Merge - 0 -49
Human Review - 50-74
Auto Merge: 75-100



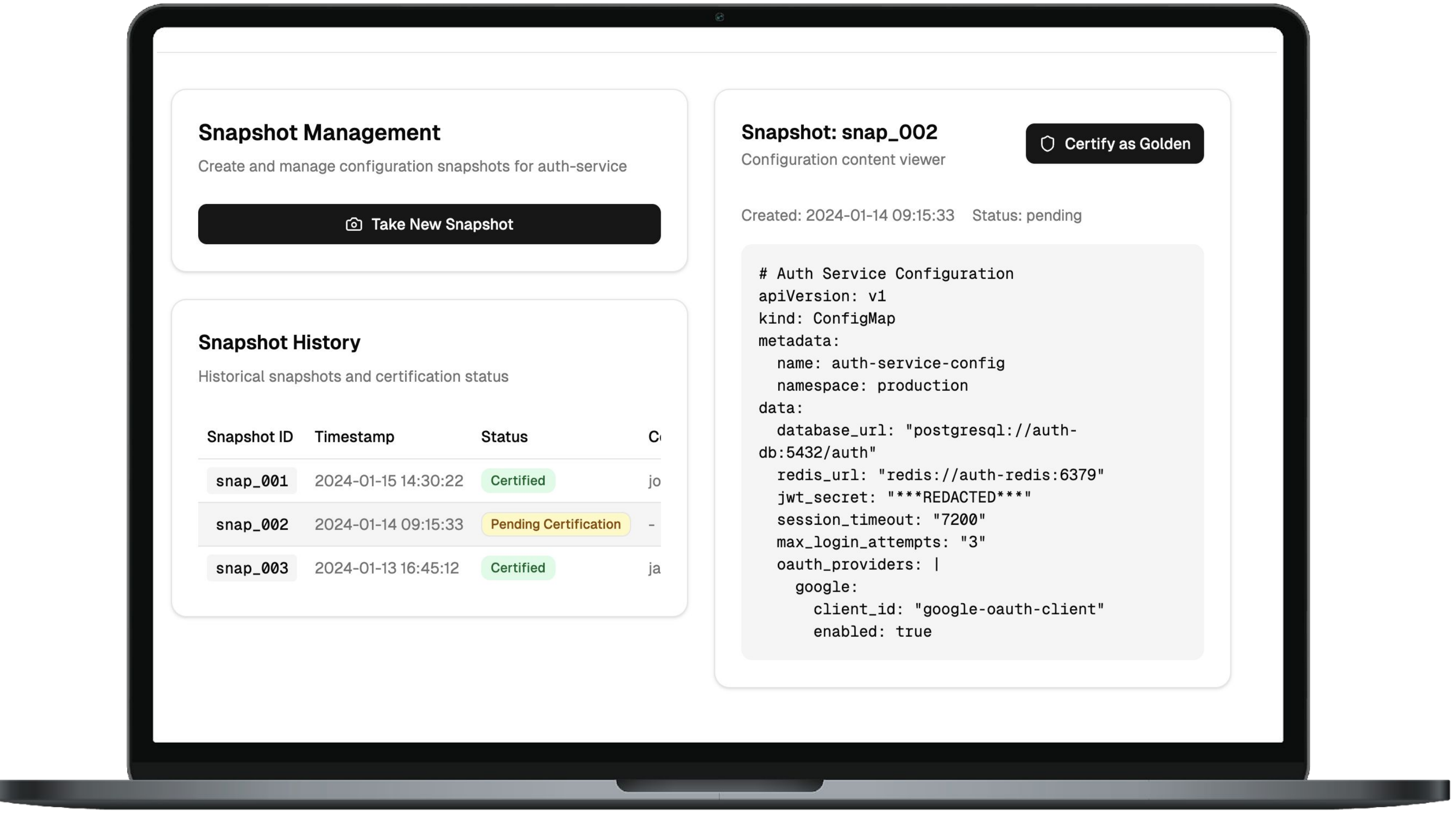
Feature Comparison AI vs Traditional

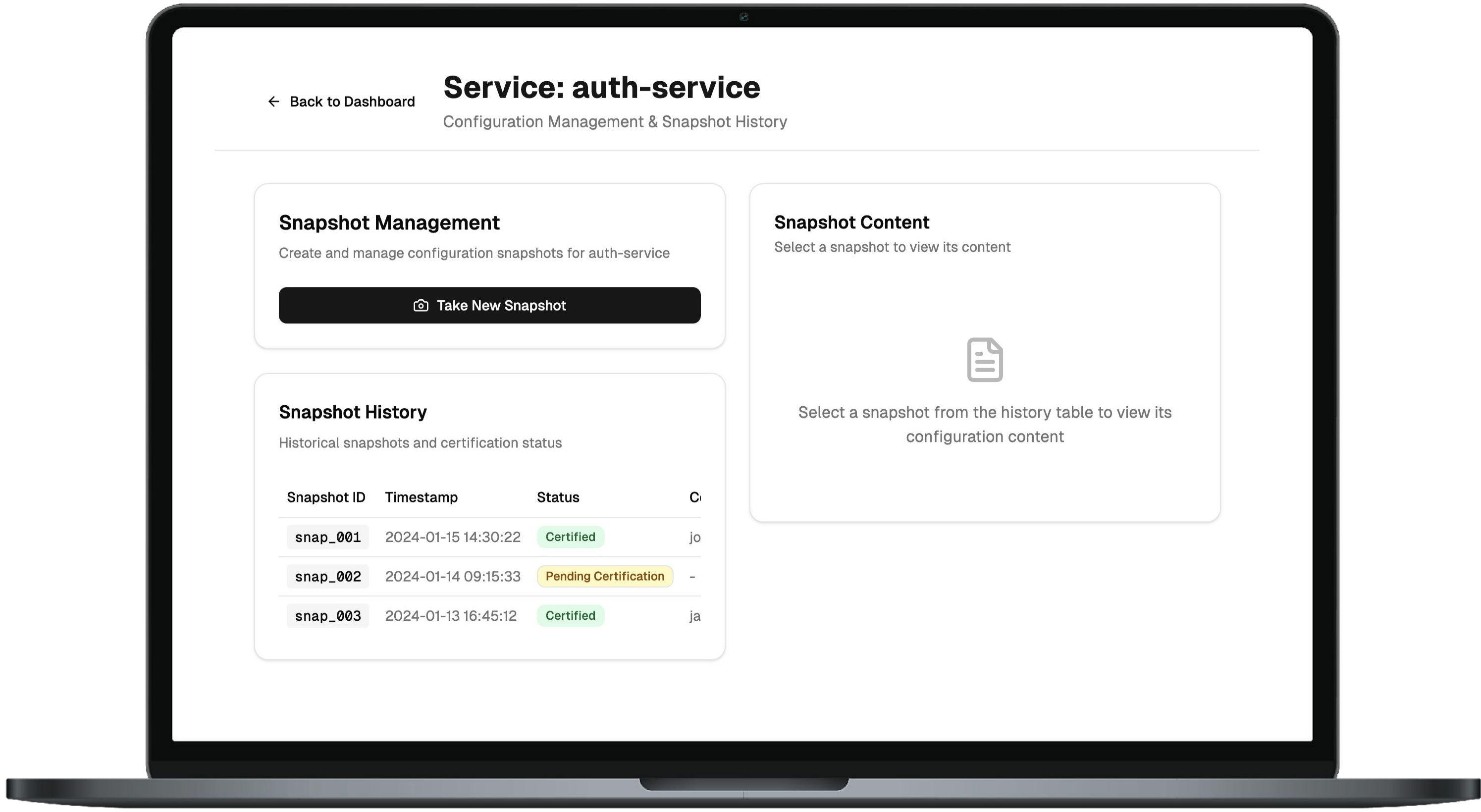
LAYER	RULES “BRAIN”	LLM “BRAIN”
Core logic	Deterministic pattern + threshold matching	Probabilistic reasoning + semantic understanding
Data scope	Single file / known policies	Full MR context, history, and text
Memory	Static rulebase	Vector-based memory across repos
Adaptivity	Manual updates	Learns patterns via few-shot retraining
Output	Binary (pass/fail)	Graded confidence with reasoning
Maintenance overhead	High (policy upkeep)	Moderate (model tuning)
Best for	Known, repeatable errors	Unknown or contextual anomalies

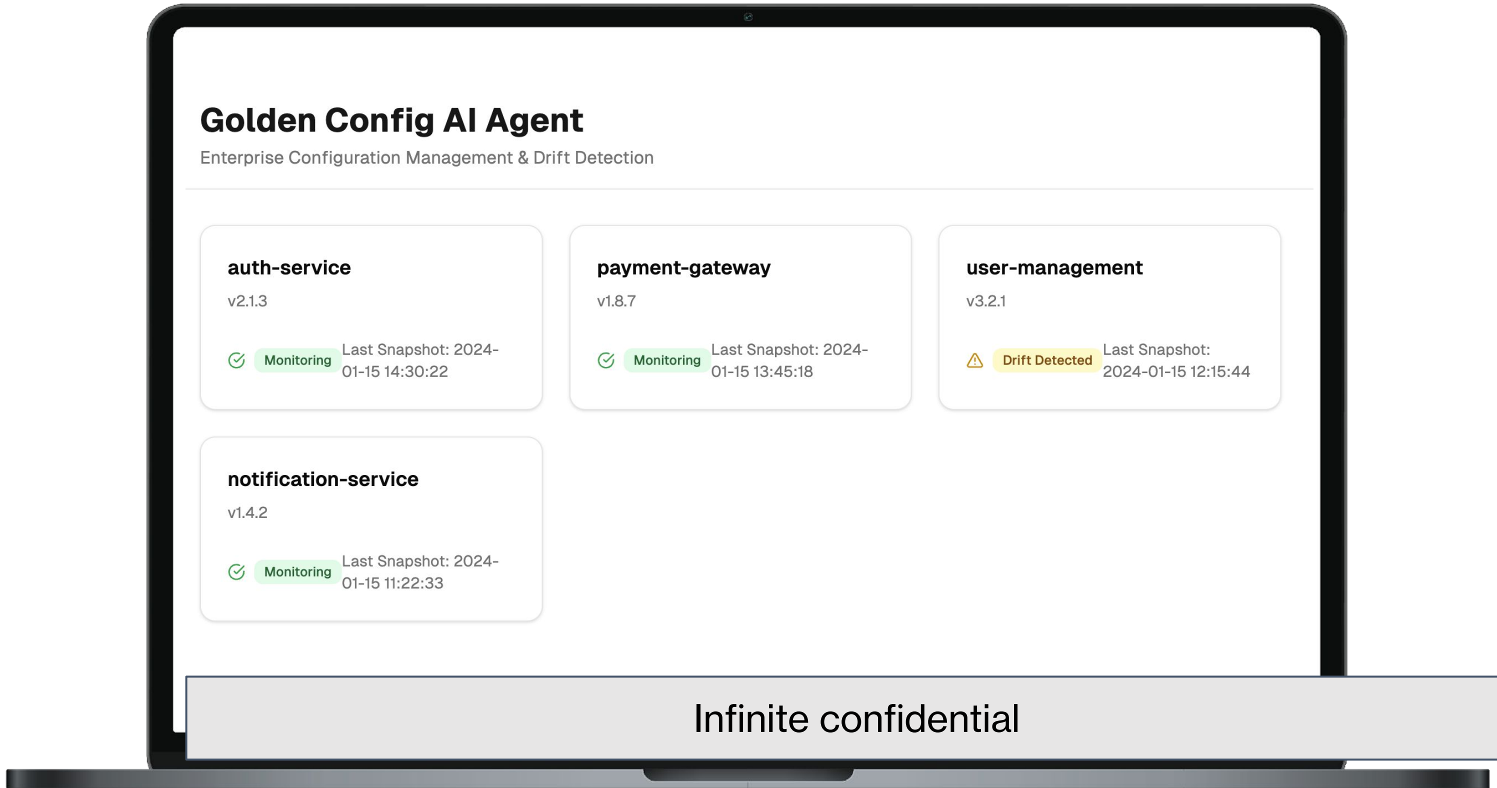
Why AI ?

Agent Decision Process (example . ptg service)
Golden: ptg-sqa3 → MR: 08v-beta2

Stage	Regex/Diff/Rules – Deterministic Brain	Bedrock LLM – Contextual Brain
1. Diff Parsing	✅ Finds line difference in `service:` field.	✅ Reads diff, MR text, and commit context together.
2. Syntax Validation	✅ Confirms value matches regex `[a-z0-9-]+`.	✅ Confirms valid syntax *and* recognizes semantic field type = service ID.
3. Static Policy Check	⚠️ Checks if service ID listed in env allowlist (if configured).	✅ Looks up known service names from text/metadata corpus; finds `08v-beta2` unknown.
4. Anomaly Detection	❌ No statistical check; treats new strings neutrally.	✅ Detects anomaly — previous envs used `ptg-sqa3`; none used `08v-beta2`.
5. Semantic Linking	❌ Doesn't cross-check downstream manifests.	✅ Connects manifests, ingress, and env mappings; sees inconsistency.
6. Contextual Inference	❌ N/A – rules can't interpret meaning.	✅ Infers likely typo from naming convention ("ptg-sqa3" pattern series).
7. Confidence Score	❌ Static scoring concept.	✅ Combines anomaly score (0.6) × rule-confidence (0.7) × text alignment (0.8) → Final 0.34 → BLOCK.
8. Decision Generation	❌ "Value changed" message only.	✅ Returns: *"Service alias `08v-beta2` not registered in Beta2; likely typo for `ptg-sqa3`. Merge blocked."*


Snapshot Management
Create and manage configuration snapshots for auth-service
Take New Snapshot
Snapshot History
Historical snapshots and certification status
Snapshot ID | Timestamp | Status
snap_001 2024-01-15 14:30:22 Certified jo
snap_002 2024-01-14 09:15:33 Pending Certification -
snap_003 2024-01-13 16:45:12 Certified ja
Snapshot: snap_002
Configuration content viewer
Certify as Golden
Created: 2024-01-14 09:15:33 Status: pending
Auth Service Configuration
apiVersion: v1
kind: ConfigMap
metadata:
name: auth-service-config
namespace: production
data:
database_url: "postgresql://auth-db:5432/auth"
redis_url: "redis://auth-redis:6379"
jwt_secret: "***REDACTED***"
session_timeout: "7200"
max_login_attempts: "3"
oauth_providers: |
google:
client_id: "google-oauth-client"
enabled: true


Back to Dashboard
Service: auth-service
Configuration Management & Snapshot History
Snapshot Management
Create and manage configuration snapshots for auth-service
Take New Snapshot
Snapshot History
Historical snapshots and certification status
Snapshot ID | Timestamp | Status
snap_001 2024-01-15 14:30:22 Certified jo
snap_002 2024-01-14 09:15:33 Pending Certification -
snap_003 2024-01-13 16:45:12 Certified ja
Snapshot Content
Select a snapshot to view its content
Select a snapshot from the history table to view its configuration content


Golden Config AI Agent
Enterprise Configuration Management & Drift Detection
auth-service
v2.1.3
Monitoring Last Snapshot: 2024-01-15 14:30:22
payment-gateway
v1.8.7
Monitoring Last Snapshot: 2024-01-15 13:45:18
user-management
v3.2.1
Drift Detected Last Snapshot: 2024-01-15 12:15:44
notification-service
v1.4.2
Monitoring Last Snapshot: 2024-01-15 11:22:33